① Information-Density-Aware Temporal Representation Reorganization

② Frequency-Selective Structured Token Interaction

③ Horizon-Conditioned Structured Extrapolation

Input

Multivariate Time Series

Value

Time

Low

High

1) Multi-Resolution Decomposition

2) Hierarchical Token Construction

Level 0 (Coarse)

Level 1 (Medium)

Level k (Fine)

Position & Level Embedding

3) Hierarchical Token Organization

Output

Value

Time

Frequency Band Partition

Low-freq

Mid-freq

High-freq

Band-wise Token Mixing

Cross-Band Fusion

Feed-Forward Network

Horizon Embedding

h_1 h_2 h_i

Horizon-Conditioned Gating

Level 0 (Coarse)

Level 1 (Medium)

Level k (Fine)

Dynamic Modeling (High Expressiveness)

Structured Extrapolation (Low Complexity Bias)

History

Future

Prediction Head

Information Density
Local complexity and informativeness

Hierarchical multi-resolution tokens

Spectral Selection
Learnable band-wise frequency selection

Structured Interaction
Efficient token mixing with cross-band fusion

Element-wise Add

Matrix Multiplication

Sigmoid Gating

Inactive/Padded Token